

Josh Muniga

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EDUCATION Bachelor of Science in Electrical Engineering - Computer Engineering May 2025

Ingram School of Engineering – ABET Accredited Program

Texas State University – San Marcos, Texas

AWARDS/HONORS

• Dean's List

Multiple Semesters

RELEVANT COURSEWORK

- VLSI Design • Data Structures & Algorithms
- Electronics 1 & 2 • Digital Signal Processing
- Linear Control Systems • Microprocessors

TECHNICAL SKILLS AND CERTIFICATIONS

- Software – Microsoft: Excel, Word, Power Point; MATLAB; Altium; LTSPICE; Fusion360; Xilinx;
- Languages/OS/Platforms - C++; Embedded C; Python; Unix/BASH; Linux; ARM, HDL, Verilog
- Written and Verbal Communication Skills

PROJECTS

NASA-Sponsored Radiation-Hardened Motherboard Development August 2024 – Present

- Designed and implemented audio schematic and MIPI camera interface using FPGA IP blocks, mapping specific BGA balls for functionality.
- Developed custom PCBs, integrating FPGA, audio, and camera components for seamless operation.
- Configured Yocto Project Linux distribution, recompiled the Linux kernel, and modified the device tree to support hardware integrations.
- Enhanced system performance through precise device configuration and low-level software optimization.
- Collaborated with a multidisciplinary team, contributing to the successful demonstration of a cutting-edge embedded system.

Design and Tuning of a PID Controller January 2024 – May 2024

- Designed and tuned a PID controller to meet specific percent overshoot (%OS) and settling time requirements for a dynamic system.
- Implemented control algorithms and conducted parameter optimization to achieve precise system control
- Ensured Compliance with performance targets while maintaining system stability and minimizing overshoot

Gyroscope System on RealDigital Blackboard and FPGA October 2023

- Developed a gyroscope using the RealDigital Blackboard microcontroller platform and FPGA for enhanced data processing
- Interfaced with an inertial measurement module for motion tracking and data acquisition.
- Utilized UART communication to design a real-time data console for efficient data transmission and monitoring.
- Optimized system for real-time processing of gyroscopic data with low latency.

EXPERIENCE Undergraduate Teaching Assistant

August 2024 – Present

Linear Control Systems , Ingram School of Engineering, Texas State University, San Marcos, TX

- Assisted the professor in delivering lectures and managing coursework for a class of 31 students • Prepared and graded assignments and projects, ensuring timely and constructive feedback to enhance student learning.
- Conducted review sessions and office hours to clarify complex concepts for students.

Server Trainer

January 2022 – Present

Cracker Barrel, San Marcos, TX

ACTIVITIES**Member,** Institute of Electrical and Electronics Engineers (IEEE)

March 2022 – Present